	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	**Aim:** LSTM	
Experiment No: 5	**Date:**	**Enrolment No:92301733057**

In [ ]:
```python
# 1.    Load the basic libraries and packages

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
from keras.models import Sequential
from keras.layers import LSTM, Dropout, Dense
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

warnings.filterwarnings('ignore')
```

In [ ]:
```python
# 2.    Load the dataset

dataset = pd.read_excel("/content/yahoo_data.xlsx")
print(dataset.head())
```

```
...            Date      Open      High       Low    Close*  Adj Close**  \
0  Apr 28, 2023  33797.43  34104.56  33728.40  34098.16     34098.16
1  Apr 27, 2023  33381.66  33859.75  33374.65  33826.16     33826.16
2  Apr 26, 2023  33596.34  33645.83  33235.85  33301.87     33301.87
3  Apr 25, 2023  33828.34  33875.49  33525.39  33530.83     33530.83
4  Apr 24, 2023  33805.04  33891.15  33726.09  33875.40     33875.40

       Volume
0  354310000
1  343240000
2  321170000
3  297880000
4  252020000
```

In [ ]:
```python
# 3.    Analyse the dataset

# Summary statistics
print(dataset.describe())

# Dataset info
print(dataset.info())
```

```
                Open          High           Low        Close*   Adj Close**  \
count   1258.000000   1258.000000   1258.000000   1258.000000   1258.000000
mean   29595.823045  29776.945739  29402.432226  29599.361677  29599.361677
std     4006.078299   4009.007573   4004.949066   4007.468822   4007.468822
min    19028.360000  19121.010000  18213.650000  18591.930000  18591.930000
25%    26041.267500  26163.155000  25877.872500  26027.120000  26027.120000
50%    29201.410000  29335.685000  28996.500000  29199.460000  29199.460000
75%    33604.027500  33825.445000  33346.827500  33600.342500  33600.342500
max    36722.600000  36952.650000  36636.000000  36799.650000  36799.650000

             Volume
count  1.258000e+03
mean   3.450636e+08
std    1.069142e+08
min    8.615000e+07
25%    2.773125e+08
50%    3.247250e+08
75%    3.875100e+08
max    9.159900e+08
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1258 entries, 0 to 1257
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         1258 non-null   object
 1   Open         1258 non-null   float64
 2   High         1258 non-null   float64
 3   Low          1258 non-null   float64
 4   Close*       1258 non-null   float64
 5   Adj Close**  1258 non-null   float64
```

In [ ]:

```python
# 4.    Apply LSTM Model

# Extract the feature column for modeling
apple_training = dataset.iloc[:, 1:2].values

# Normalize the feature using Min-Max Scaler
scaler = MinMaxScaler(feature_range=(0, 1))
apple_training_scaled = scaler.fit_transform(apple_training)

# Create the feature set and labels
feature_set = []
labels = []

for i in range(60, 253):
    feature_set.append(apple_training_scaled[i-60:i,  0])
    labels.append(apple_training_scaled[i,  0])

# Convert to numpy arrays
feature_set = np.array(feature_set)
labels = np.array(labels)

# Reshape feature set for LSTM input
feature_set = np.reshape(feature_set, (feature_set.shape[0], feature_set.shape[1

# Initialize the model
```

```
model = Sequential()

# Add LSTM layers with Dropout
model.add(LSTM(units=60, return_sequences=True, input_shape=(feature_set.shape[1
model.add(Dropout(0.20))

model.add(LSTM(units=60, return_sequences=True))
model.add(Dropout(0.20))

model.add(LSTM(units=60, return_sequences=True))
model.add(Dropout(0.20))

model.add(LSTM(units=60))
model.add(Dropout(0.20))

# Add output layer
model.add(Dense(units=1))

# Compile the model
model.compile(optimizer='adam',  loss='mean_squared_error',  metrics=['accuracy'])
```

In [ ]:

```
# 5.    Apply the training over the dataset to minimize the loss

history = model.fit(feature_set, labels, epochs=100, batch_size=32, validation_s
```

```
Epoch 1/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 9s 306ms/step - accuracy: 0.0000e+00 - loss: 0.2080 - va
l_accuracy: 0.0256 - val_loss: 0.1115
Epoch 2/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 2s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0283 - va
l_accuracy: 0.0256 - val_loss: 0.0404
Epoch 3/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0179 - va
l_accuracy: 0.0256 - val_loss: 0.0407
Epoch 4/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 142ms/step - accuracy: 0.0000e+00 - loss: 0.0105 - va
l_accuracy: 0.0256 - val_loss: 0.0950
Epoch 5/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0083 - va
l_accuracy: 0.0256 - val_loss: 0.0503
Epoch 6/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 143ms/step - accuracy: 0.0000e+00 - loss: 0.0063 - va
l_accuracy: 0.0256 - val_loss: 0.1485
Epoch 7/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 209ms/step - accuracy: 0.0000e+00 - loss: 0.0044 - va
l_accuracy: 0.0256 - val_loss: 0.0979
Epoch 8/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 238ms/step - accuracy: 0.0000e+00 - loss: 0.0047 - va
l_accuracy: 0.0256 - val_loss: 0.1641
Epoch 9/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 243ms/step - accuracy: 0.0000e+00 - loss: 0.0041 - va
l_accuracy: 0.0256 - val_loss: 0.1058
Epoch 10/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 171ms/step - accuracy: 0.0000e+00 - loss: 0.0044 - va
l_accuracy: 0.0256 - val_loss: 0.1282
Epoch 11/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0050 - va
l_accuracy: 0.0256 - val_loss: 0.1112
Epoch 12/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 133ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.0834
Epoch 13/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0048 - va
l_accuracy: 0.0256 - val_loss: 0.1075
Epoch 14/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 160ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.1137
Epoch 15/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 147ms/step - accuracy: 0.0000e+00 - loss: 0.0044 - va
l_accuracy: 0.0256 - val_loss: 0.1124
Epoch 16/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0032 - va
l_accuracy: 0.0256 - val_loss: 0.1214
Epoch 17/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0043 - va
l_accuracy: 0.0256 - val_loss: 0.1325
Epoch 18/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0040 - va
l_accuracy: 0.0256 - val_loss: 0.1257
Epoch 19/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 137ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.1298
Epoch 20/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 2s 254ms/step - accuracy: 0.0000e+00 - loss: 0.0044 - va
l_accuracy: 0.0256 - val_loss: 0.1019
```

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Epoch 21/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 217ms/step - accuracy: 0.0000e+00 - loss: 0.0051 - va
l_accuracy: 0.0256 - val_loss: 0.1289
Epoch 22/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 253ms/step - accuracy: 0.0000e+00 - loss: 0.0038 - va
l_accuracy: 0.0256 - val_loss: 0.1206
Epoch 23/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 145ms/step - accuracy: 0.0000e+00 - loss: 0.0037 - va
l_accuracy: 0.0256 - val_loss: 0.1095
Epoch 24/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 135ms/step - accuracy: 0.0000e+00 - loss: 0.0032 - va
l_accuracy: 0.0256 - val_loss: 0.1175
Epoch 25/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.1151
Epoch 26/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 150ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.1534
Epoch 27/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 143ms/step - accuracy: 0.0000e+00 - loss: 0.0048 - va
l_accuracy: 0.0256 - val_loss: 0.0952
Epoch 28/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 135ms/step - accuracy: 0.0000e+00 - loss: 0.0042 - va
l_accuracy: 0.0256 - val_loss: 0.0996
Epoch 29/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 136ms/step - accuracy: 0.0000e+00 - loss: 0.0053 - va
l_accuracy: 0.0256 - val_loss: 0.1661
Epoch 30/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 140ms/step - accuracy: 0.0000e+00 - loss: 0.0057 - va
l_accuracy: 0.0256 - val_loss: 0.0771
Epoch 31/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 154ms/step - accuracy: 0.0000e+00 - loss: 0.0041 - va
l_accuracy: 0.0256 - val_loss: 0.1290
Epoch 32/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 166ms/step - accuracy: 0.0000e+00 - loss: 0.0032 - va
l_accuracy: 0.0256 - val_loss: 0.1154
Epoch 33/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 220ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.0862
Epoch 34/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 242ms/step - accuracy: 0.0000e+00 - loss: 0.0048 - va
l_accuracy: 0.0256 - val_loss: 0.1131
Epoch 35/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 170ms/step - accuracy: 0.0000e+00 - loss: 0.0036 - va
l_accuracy: 0.0256 - val_loss: 0.0953
Epoch 36/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0039 - va
l_accuracy: 0.0256 - val_loss: 0.1399
Epoch 37/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0042 - va
l_accuracy: 0.0256 - val_loss: 0.0687
Epoch 38/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 157ms/step - accuracy: 0.0000e+00 - loss: 0.0054 - va
l_accuracy: 0.0256 - val_loss: 0.1571
Epoch 39/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 152ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.0882
Epoch 40/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 152ms/step - accuracy: 0.0000e+00 - loss: 0.0039 - va
l_accuracy: 0.0256 - val_loss: 0.1360
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Epoch 41/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 154ms/step - accuracy: 0.0000e+00 - loss: 0.0041 - va
l_accuracy: 0.0256 - val_loss: 0.1083
Epoch 42/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0036 - va
l_accuracy: 0.0256 - val_loss: 0.1094
Epoch 43/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 2s 284ms/step - accuracy: 0.0000e+00 - loss: 0.0040 - va
l_accuracy: 0.0256 - val_loss: 0.1059
Epoch 44/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 155ms/step - accuracy: 0.0000e+00 - loss: 0.0038 - va
l_accuracy: 0.0256 - val_loss: 0.1056
Epoch 45/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 215ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.1033
Epoch 46/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 218ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.1380
Epoch 47/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 238ms/step - accuracy: 0.0000e+00 - loss: 0.0032 - va
l_accuracy: 0.0256 - val_loss: 0.0830
Epoch 48/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 154ms/step - accuracy: 0.0000e+00 - loss: 0.0032 - va
l_accuracy: 0.0256 - val_loss: 0.1023
Epoch 49/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 171ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.1297
Epoch 50/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 148ms/step - accuracy: 0.0000e+00 - loss: 0.0037 - va
l_accuracy: 0.0256 - val_loss: 0.0893
Epoch 51/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 140ms/step - accuracy: 0.0000e+00 - loss: 0.0032 - va
l_accuracy: 0.0256 - val_loss: 0.1102
Epoch 52/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 142ms/step - accuracy: 0.0000e+00 - loss: 0.0028 - va
l_accuracy: 0.0256 - val_loss: 0.0968
Epoch 53/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 135ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.0889
Epoch 54/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0029 - va
l_accuracy: 0.0256 - val_loss: 0.1324
Epoch 55/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 154ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.0828
Epoch 56/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 137ms/step - accuracy: 0.0000e+00 - loss: 0.0036 - va
l_accuracy: 0.0256 - val_loss: 0.1027
Epoch 57/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 170ms/step - accuracy: 0.0000e+00 - loss: 0.0036 - va
l_accuracy: 0.0256 - val_loss: 0.1060
Epoch 58/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 2s 223ms/step - accuracy: 0.0000e+00 - loss: 0.0036 - va
l_accuracy: 0.0256 - val_loss: 0.0803
Epoch 59/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 240ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.1064
Epoch 60/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.0682
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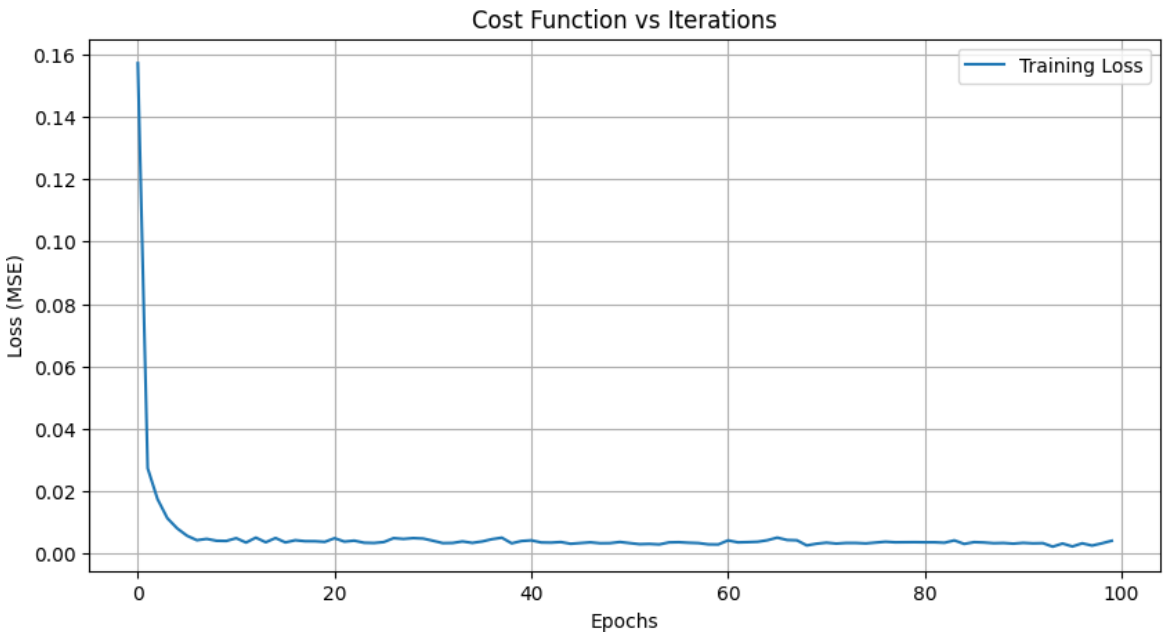
Epoch 61/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 158ms/step - accuracy: 0.0000e+00 - loss: 0.0040 - va
l_accuracy: 0.0256 - val_loss: 0.1269
Epoch 62/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 180ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.0717
Epoch 63/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 155ms/step - accuracy: 0.0000e+00 - loss: 0.0037 - va
l_accuracy: 0.0256 - val_loss: 0.1005
Epoch 64/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 172ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.1071
Epoch 65/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 149ms/step - accuracy: 0.0000e+00 - loss: 0.0041 - va
l_accuracy: 0.0256 - val_loss: 0.0611
Epoch 66/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 148ms/step - accuracy: 0.0000e+00 - loss: 0.0062 - va
l_accuracy: 0.0256 - val_loss: 0.1267
Epoch 67/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0047 - va
l_accuracy: 0.0256 - val_loss: 0.0668
Epoch 68/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 147ms/step - accuracy: 0.0000e+00 - loss: 0.0043 - va
l_accuracy: 0.0256 - val_loss: 0.1003
Epoch 69/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 145ms/step - accuracy: 0.0000e+00 - loss: 0.0025 - va
l_accuracy: 0.0256 - val_loss: 0.0796
Epoch 70/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 187ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.1086
Epoch 71/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 219ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.0835
Epoch 72/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 256ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.1182
Epoch 73/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 179ms/step - accuracy: 0.0000e+00 - loss: 0.0036 - va
l_accuracy: 0.0256 - val_loss: 0.0893
Epoch 74/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 143ms/step - accuracy: 0.0000e+00 - loss: 0.0037 - va
l_accuracy: 0.0256 - val_loss: 0.0843
Epoch 75/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 135ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.1086
Epoch 76/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 156ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.0901
Epoch 77/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0041 - va
l_accuracy: 0.0256 - val_loss: 0.1214
Epoch 78/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0037 - va
l_accuracy: 0.0256 - val_loss: 0.0676
Epoch 79/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0038 - va
l_accuracy: 0.0256 - val_loss: 0.1096
Epoch 80/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 147ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.0891
```

```
Epoch 81/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 146ms/step - accuracy: 0.0000e+00 - loss: 0.0039 - va
l_accuracy: 0.0256 - val_loss: 0.0996
Epoch 82/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.1006
Epoch 83/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 140ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.0709
Epoch 84/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 188ms/step - accuracy: 0.0000e+00 - loss: 0.0047 - va
l_accuracy: 0.0256 - val_loss: 0.1142
Epoch 85/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 220ms/step - accuracy: 0.0000e+00 - loss: 0.0035 - va
l_accuracy: 0.0256 - val_loss: 0.0879
Epoch 86/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 243ms/step - accuracy: 0.0000e+00 - loss: 0.0039 - va
l_accuracy: 0.0256 - val_loss: 0.1092
Epoch 87/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 138ms/step - accuracy: 0.0000e+00 - loss: 0.0037 - va
l_accuracy: 0.0256 - val_loss: 0.0865
Epoch 88/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 154ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.1035
Epoch 89/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.0873
Epoch 90/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 159ms/step - accuracy: 0.0000e+00 - loss: 0.0030 - va
l_accuracy: 0.0256 - val_loss: 0.1020
Epoch 91/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 141ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.0784
Epoch 92/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 146ms/step - accuracy: 0.0000e+00 - loss: 0.0034 - va
l_accuracy: 0.0256 - val_loss: 0.1238
Epoch 93/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 153ms/step - accuracy: 0.0000e+00 - loss: 0.0033 - va
l_accuracy: 0.0256 - val_loss: 0.0985
Epoch 94/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 143ms/step - accuracy: 0.0000e+00 - loss: 0.0022 - va
l_accuracy: 0.0256 - val_loss: 0.0951
Epoch 95/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 140ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.1170
Epoch 96/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 175ms/step - accuracy: 0.0000e+00 - loss: 0.0026 - va
l_accuracy: 0.0256 - val_loss: 0.0748
Epoch 97/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 2s 245ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.1113
Epoch 98/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 244ms/step - accuracy: 0.0000e+00 - loss: 0.0027 - va
l_accuracy: 0.0256 - val_loss: 0.0744
Epoch 99/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 134ms/step - accuracy: 0.0000e+00 - loss: 0.0031 - va
l_accuracy: 0.0256 - val_loss: 0.1246
Epoch 100/100
5/5 ━━━━━━━━━━━━━━━━━━━━ 1s 139ms/step - accuracy: 0.0000e+00 - loss: 0.0041 - va
l_accuracy: 0.0256 - val_loss: 0.0660
```

In [ ]:
```
# 6.    Observe the cost function vs iterations learning curve

# Plot cost function (loss) vs iterations (epochs)
plt.figure(figsize=(10, 5))
plt.plot(history.history['loss'], label='Training Loss')
plt.title('Cost Function vs Iterations')
plt.xlabel('Epochs')
plt.ylabel('Loss (MSE)')
plt.legend()
plt.grid(True)
plt.show()
```



Result

In [ ]:
```
# a.    Model Summary

model.summary()
```

Model: "sequential_1"

Layer (type)	Output Shape
lstm_4 (LSTM)	(None, 60, 60)
dropout_4 (Dropout)	(None, 60, 60)
lstm_5 (LSTM)	(None, 60, 60)
dropout_5 (Dropout)	(None, 60, 60)
lstm_6 (LSTM)	(None, 60, 60)
dropout_6 (Dropout)	(None, 60, 60)
lstm_7 (LSTM)	(None, 60)
dropout_7 (Dropout)	(None, 60)
dense_1 (Dense)	(None, 1)

Total params: 306,185 (1.17 MB)

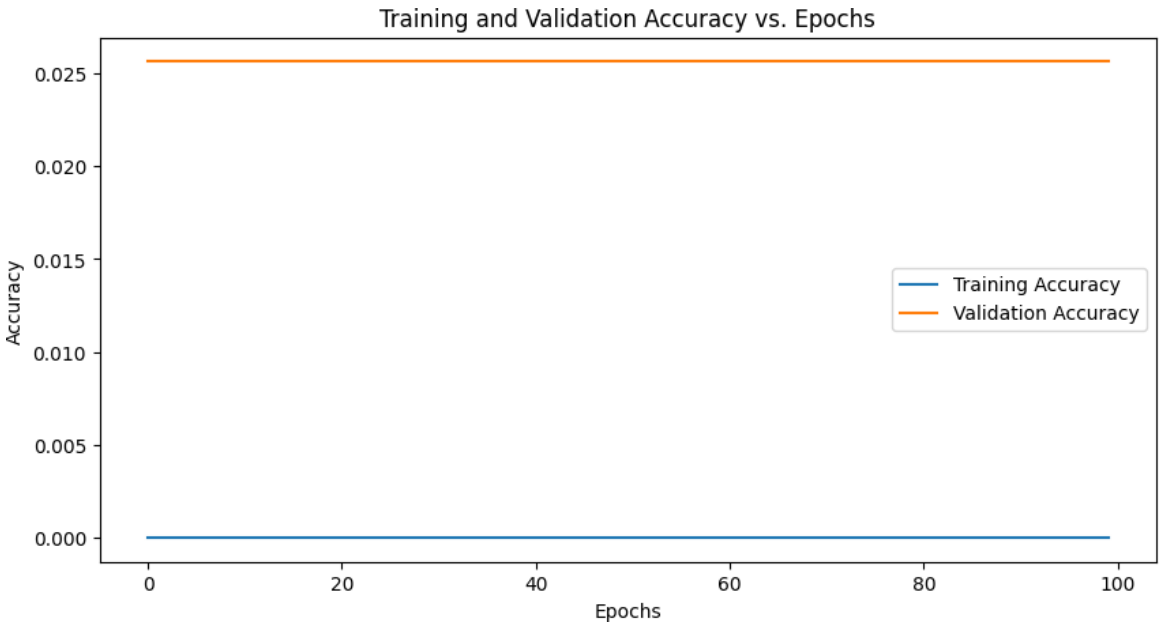
Trainable params: 102,061 (398.68 KB)

Non-trainable params: 0 (0.00 B)

Optimizer params: 204,124 (797.36 KB)

In [ ]:
```
# b.    Training and Validation accuracy v/s epochs

# Plot training and validation accuracy
plt.figure(figsize=(10, 5))
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Training and Validation Accuracy vs. Epochs')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
plt.show()
```



In [ ]:
```
# c.    Training and Validation loss v/s epochs

# Plot training and validation loss
plt.figure(figsize=(10, 5))
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Training and Validation Loss vs. Epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()
plt.show()
```

In [ ]:
```
# d. Visualize the Predicted and originalStock Price

# Prepare testing data
testing_data = dataset.iloc[:, 1:2].values
test_feature = testing_data.reshape(-1, 1)
test_feature = scaler.transform(test_feature)

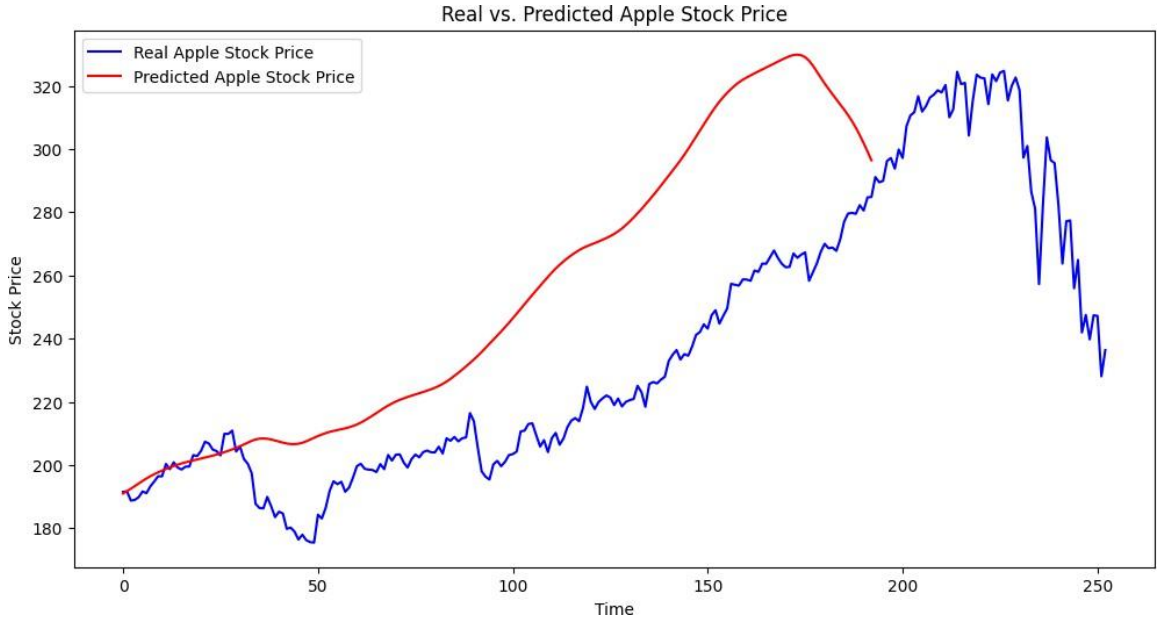
testing_features = []

for i in range(60, 253):
    testing_features.append(test_feature[i-60:i, 0])

testing_features = np.array(testing_features)
testing_features = np.reshape(testing_features, (testing_features.shape[0], test

# Predict the stock prices
predictions = model.predict(testing_features)
predictions = scaler.inverse_transform(predictions)

# Visualize the results
plt.figure(figsize=(12, 6))
plt.plot(testing_data, color='blue', label='Real Apple Stock Price')
plt.plot(predictions, color='red', label='Predicted Apple Stock Price')
plt.title('Real vs. Predicted Apple Stock Price')
plt.xlabel('Time')
plt.ylabel('Stock Price')
```

```
plt.legend()
plt.show()
```

```
7/7 ━━━━━━━━━━━━━━━━━━━━ 2s 287ms/step
```